



Project presentation

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Optimizing the CS catalog

PEAS Assessment:

- Performance Measure: Graduation rate (semesters).
- Environment: Fully observable, deterministic, static, discrete..
- Actuators: A student taking classes from the catalog.
- Sensors: Available courses in terms of semester and prerequisites.



Project Design

CATALOG RUNNER:

01.

CATALOG SEARCH

By running “do_catalog_search” you can find the best catalog to then use if with the agents.

You can change some parameters: random restarts or number of students.

02.

CATALOG MEASURE

By running “do_catalog_measure” you can measure how good a catalog is with the different agents.

You can change some parameters: credits per semester and number of students.

Problem Instances and Results

CATALOG SEARCH

Restarts	Students	Utility (semesters)	Run Time (seconds)
5	100	9.63	5.80
10	100	9.68	11.15
50	100	9.69	56.30
5	1,000	9.74	50.06
10	1,000	9.70	114.44
50	1,000	9.74	565.90
10	10,000	9.72	1105.58

RANDOM AGENT

Credits	Av. Semesters	Min Semester	Max Semester	Students
12	15.00	12	20	100
15	12.00	10	17	100
18	11.00	8	14	100
20	10.00	7	14	100

Credits	Av. Semesters	Min Semester	Max Semester	Students
12	14.00	10	21	1,000
15	12.00	9	18	1,000
18	11.00	7	16	1,000
20	10.00	8	15	1,000

REQUIREMENTS AGENT

Credits	Av. Semesters	Min Semester	Max Semester	Students
12	8.00	7	10	100
15	6.00	6	8	100
18	5.00	5	6	100
20	5.00	5	6	100

Credits	Av. Semesters	Min Semester	Max Semester	Students
12	7.00	7	10	1,000
15	6.00	6	8	1,000
18	5.00	5	6	1,000
20	5.00	5	6	1,000

Future Improvements

01

Make an agent that takes into account if too many hard classes are taken together.

02

Add the general classes into the catalog for a better simulation of students.

03

Implementing a system that checks not just immediate prerequisites but also prerequisite chains.

04

Incorporate real-world data on course offerings, student performance, and graduation rates to make the simulation more realistic.

The background is a light gray grid. It is decorated with various hand-drawn blue doodles. At the top, there are several overlapping circles and loops. On the right side, there are some star-like or burst-like shapes. At the bottom, there are more circles, a wavy line, and several small checkmarks or 'v' shapes.

**Thank you
very much!**